Optimal Stratified Sampling for Probability-Based Online Panels

by

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Abstract

Online probability-based panels have emerged as a cost-efficient means of conducting surveys in the 21st century. While there have been various recent advancements in sampling techniques for online panels, several critical aspects of sampling theory for online panels are lacking. Much of current sampling theory from the middle of the 20th century, when response rates were high, and online panels did not exist. This paper presents a mathematical model of stratified sampling for online panels that takes into account historical response rates and survey costs. Through some simplifying assumptions, the model shows that the optimal sample allocation for online panels can largely resemble the solution for a cross-sectional survey. To apply the model, I use the Census Household Panel to show how this method could improve the average precision of key estimates. Holding fielding costs constant, the new sample rates improve the average precision of estimates between 1.47 and 17.25 percent, depending on the importance weight given to an overall population mean compared to mean estimates for racial and ethnic subgroups.

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1. Introduction

In the current internet age, online probability-based panels have emerged as a popular means of conducting surveys given the cost-savings benefits of having an internet response mode combined with the theoretical justifications of probability-based sampling. Examples of such panels in the United States include NORC's AmeriSpeak, Pew's American Trends Panel, Ipsos's KnowledgePanel, and the recent Household Trends and Outlook Pulse Survey (HTOPS) launched by the U.S. Census Bureau in 2025. These panels use address-based sampling to recruit households by sending them invitations in the mail, followed by sending survey notifications primarily via email and text once they are past the recruitment stage.

There have also been recent advancements in the ability of such panels to conduct stratified sampling, which has the potential to improve the precision of estimates. In the past, the information known about a household with address-based sampling has largely been limited to geographic information. Using only geographic information for design sampling strata may result in sampling strata that are only weakly correlated with key survey outcomes of interest. Recently, there has been an increase in commercial data on individuals, allowing for sampling strata definitions that have a higher correlation with survey outcomes of interest. AmeriSpeak uses such data to stratify the sample by age, race-ethnicity, and political affiliation (NORC 2024). In the government sector, increases in data harmonization and linkage abilities have led to the increased use of government records in stratified sampling strategies. HTOPS uses a harmonized dataset called the Demographic Frame (Ratcliffe 2021) to oversample households containing non-White and Hispanic individuals. Advances in machine-learning models also allow for combining multiple sources to create sampling-strata that are correlated well with survey outcomes. Dutwin et al. (2024) use various machine-learning models to create sampling strata for the AmeriSpeak panel; Eggleston et al. (2025) use random forest classifiers to develop sampling strata for the FoodAPS Pilot Survey.

While there have been numerous advances in creating sampling strata for surveys, one area that has not received as much recent attention is how to best pick sampling rates for stratified samples. Much of current sampling theory is from texts published earlier in the 20th century, such as Cochran (1977) and Kalton and Anderson (1986), which assume 100% response rates in their sampling rate calculations. There have been some recent advances for more modern settings, such as Mendelson and Elliott (2024) which present analytical solutions for stratified sampling rates for cross-sectional surveys with incomplete response. Additionally, there have been some applications of numerical optimization methods for constructing sampling rates, such as Valliant et al. (1997, 2014) and Gomes and Johnson (2016). While Valliant et al. (2014) did apply their model to the Health and Retirement Study, a longitudinal survey, the framework presented in these prior papers was that of a cross-sectional survey.

The lack of explicit sampling solutions for online panels or longitudinal surveys poses difficulties for survey practitioners. While one can apply the formula for a cross-sectional survey to an online panel, it is unclear how appropriate this application would be. For example, a key determinate of sampling rates in formulas from Cochran (1977) is the cost per case difference between the strata. For online panels, these costs include the costs of the recruitment survey and of conducting subsequent topical surveys. The recruitment costs have the potential to be larger if mail panel invitations are primarily used for recruitment, but then invitations for topical surveys are sent out through email or texting. Given that formulas from Cochran (1977) and Mendelson and Elliott (2024) don't have such distinctions, it's unclear

how best to assign cost parameters in order to apply cross-sectional sampling rate formulas for online panels.

This paper fills in the gap in the literature by presenting a mathematical model for optimal sample allocation for online panel surveys. The focus of this paper is how to design sampling rates for the initial recruitment stage rather than the topical surveys, which is a modeling choice I make in order to reduce the dimensionality of the sampling problem. The goal of this model is to strike a balance between adding various features relevant to online panels, but also to make some simplifying assumptions in order to keep the model tractable. The model presented in this paper accommodates differential costs and response rates between the recruitment survey and the topical surveys, reflecting key features of online panels. However, it relies on simplifying assumptions, such as a constant sampling rate for topical surveys within each stratum and the absence of panel attrition. In the end, the model I present largely resembles the model for a cross-sectional survey, but with a more complicated cost function that includes additional cost parameters. Numerical optimization methods are used to solve for this model. Full details of the numerical optimization methods are described in Appendix Section A.1.

To apply this model, I use the Census Household Panel (CHP), a predecessor to the U.S. Census Bureau's Household Trends and Outlook Pulse Survey (HTOPS). The key results in this application are how the precision of estimates change when using the new optimal sampling rates. The outcomes of interest are both the overall population mean as well as means for race and ethnicity subgroups. The estimates' improvements vary depending on how one prioritizes precision of the overall national estimate compared to the subgroup analyses. Using a specification that results in sampling rates similar to the original design, the new optimal sampling rates result in an average decrease in precision of 1.69% for the same fielding cost. Alternative specifications that place higher importance on subgroup analyses result in more distinct sampling rates. In one of these specifications, the average precision of estimates is decreased by 15.01%. Expressing this change in precision in an alternative cost-saving format, the use of optimal sampling rates could result in having the same precision of estimates for a budget that is 16.00% percent less.

2. Model

Consider an online panel survey, where the decision at hand for sampling is deciding how many sampling units to sample from various strata for the initial recruitment. Like many other sampling problems, I assume all the other design features (e.g. number of mailings to send) are fixed, so the sample sizes are the only variables left to decide. For this online panel design specifically, the specific sample size they choose is the number of cases to sample for the initial panel recruitment efforts. After this choice, the number of panelists they recruit as well as the number of individuals who respond to the topical surveys are random variables they do not control. In this paper, I approach the sampling problem from a finite population perspective, where the characteristics of households and individuals is fixed ahead of time. Let the population be divided into H strata. N is the overall population size, and N_h is the population size for stratum h, which is assumed to be known. The only random variables are whether a population unit is sampled for the survey, and whether these sampled individuals will be respondents.

2.1 Mechanism for Number of Respondents

If an address is sampled for the panel, I assume that the probability that they are out of scope for the panel (e.g. housing unit is vacant) is ψ_h, which is constant within each stratum. If the household is in-

scope, then their stratum-constant recruitment rate is ϕ_h. Thus, if an initial sample size of n_h^I is drawn for stratum h, then the expected number of respondents is $E(n_h^R) = \phi_h(1-\psi_h)n_h^I$ if the number of respondents, nonrespondents, and ineligible cases follows a multinomial distribution. However, one problem with the multinomial distribution is the non-zero probability of the number of respondents being zero, which would lead to the variance of the estimator becoming undefined. Mendelson and Elliott (2024), which does not consider ineligible cases, solve this problem by assuming n_h^R follows a truncated binomial distribution, where n_h^R is truncated to always be greater than or equal to one. I employ the same restriction in our multinomial setting. If the initial sample or the value of $\phi_h(1-\psi_h)$ is large enough, then there is no meaningful difference between the truncated and untruncated distributions, so this assumption largely becomes a mathematical formality. Additionally, having the number of cases in the three categories follow a truncated multinomial distribution as opposed to a more complicated distribution makes the distribution easier to simulate from, which is beneficial when using numerical methods to find the solution.

Once the recruitment phase is complete, there are n_h^R panel members in each stratum. The panel members are in the panel for T years and are invited to surveys in every time period t. For each period, panelists are sampled for topical survey S with probability $\xi_{s,h}$ that is constant within stratum. The probability of responding to survey s is $\phi'_{s,h}$ which is independent of t (so there is no attrition or panel fatigue). The expected number of respondents for topic s in year t is $E(n'_{t,s,h}) = \xi_{s,h}\phi'_{s,h}n_h^R$. Note that the realized value can vary by t, even if the expectation is independent of t.

2.2 Survey Outcome and Objective Function

In the sampling problem, the ultimate outcome of interest is the precision of various estimates. For each topical survey, let there be K^s outcomes of interest, with the tuple (k, s) indexing a particular outcome. I assume these outcomes are all means with population parameters $\theta^{s,k}$, which is a finite-population average over all units. For these outcomes, let $Y_{t,h,i}^{s,k}$ denote the value of outcome s, k for household i in stratum h at time t. Within stratum h, the variable $Y_{t,h,i}^{s,k}$ is drawn from a distribution with a mean $\theta_h^{s,k}$ and variance $\sigma_h^{s,k}$. $\theta_h^{s,k}$ can be the mean for either the entire population or a domain-specific mean for a particular subgroup. To simply the problem, I assume $Y_{t,h,i}^{s,k}$ is drawn independently over time for individual i. This makes the solution more tractable but does ignore intra-individual correlation that is likely present in real-world data.

For domain specific means, the indicator variable $d_{t,h,i}^{s,k}$ equals 1 if household i is in this domain. I let the probability of a respondent being in the domain be $\delta_h^{s,k}$ which varies by stratum. With the finite population perspective, I treat domain membership as fixed ahead of time, similar to a household's value of an outcome variable. What is uncertain is whether a sampled household will end of being a member of a certain domain. The total number of respondents in this domain at time t who respond to topical survey s is denoted by $D_{t,h}^{s,k}$. Conditional on a value of n_h^R, I let $D_{t,h}^{s,k}$ be drawn from a truncated binomial distribution to ensure that $D_{t,h}^{s,k} >= 1$. For a given topical survey, the expected number of respondents used to calculate the domain-specific mean would be $\xi_{s,h}\phi'_{s,h}\delta_h^{s,k}n_h^R$.

Our estimate of $\theta^{s,k}$ is $\widehat{\theta^{s,k}}$. For each stratum, the estimate of the mean of Y conditional on being in stratum h is given by $\sum_{i=1}^{n'_{t,s,h}} Y_{t,h,i}^{s,k} d_{t,h,i}^{s,k} / D_{t,h}^{s,k}$. For non-domain means, this equation simplifies to $\sum_{i=1}^{n'_{t,s,h}} Y_{t,h,i}^{s,k} / n'_{t,s,h}$.

In our model, since the strata sizes are assumed to be known, each stratum-specific mean can be weighted by the stratum size to get the overall estimate. For domain means, what is needed is the stratum size of the domain. For these cases, let $M_h^{s,k} = \delta_h^{s,k} N_h$ is the population size for stratum h that is in the domain, and $M^{s,k} = \sum_{h=1}^{H} M_h^{s,k}$ be the total domain population size. This formula is based on $\delta_h^{s,k}$ being a known parameter, when in practice these parameters are typically unknown and need to be estimated.

Because of the assumptions on constant response rate within stratum, this is the only weighting that needs to be done to achieve unbiased estimates (for at least non-domain means). The post-stratified designed-based estimator for $\widehat{\theta^{s,k}}$ is:

$$\sum_{h=1}^{H} \left(\frac{M_h^{s,k}}{M^{s,k}} \right) \sum_{i=1}^{n'_{t,s,h}} \frac{Y_{t,h,i}^{s,k} d_{t,h,i}^{s,k}}{D_{t,h}^{s,k}} \tag{1}$$

In survey statistics, a common quality metric of an estimator is its mean squared error. Our objective function is largely in this form. Let $E\left[\left(\widehat{\theta_t^{s,k}} - \theta^{s,k}\right)^2\right]$ denote the mean squared error. Given I have multiple mean squared errors, I assign a weight to each variable to denote how important the accuracy of the statistic is relative to others. Let ω denote these weights, with $\sum_{k=1}^{K^s} \omega_{s,k} = 1$ and $\omega_{s,k} > 0$. To make the mean squared errors unitless, I divide by the population parameter squared.

Additionally, since the online panel is conducted over time, the survey practitioner cares about minimizing the weighted mean squared errors over time. To borrow concepts from economics (e.g. Arcidiacono and Ellickson 2011), I assume that at time $t = 0$, when the survey practitioner is making the initial sampling decisions, the survey practitioner cares more about the mean squared error at time t=0 compared to later years. Then let the error for each year be discounted by a factor of β^t with $0 < \beta < 1$. From this, the objective function to minimize is:

$$\min_{\{n_1^s, \ldots, n_H^s\}} \sum_{t=1}^{T} \beta^t \left(\sum_{s=1}^{S} \sum_{k=1}^{K^s} \frac{\omega_{s,k} E\left[\left(\widehat{\theta_t^{s,k}} - \theta^{s,k}\right)^2\right]}{(\theta^{s,k})^2} \right) \tag{2}$$

Subject to a cost constraint, which is discussed in the following section. Note that the term after β^t is independent of t. Using the standard annuity formula, this equation can then be reexpressed as

$$\min_{\{n_1^s,\ldots,n_H^s\}} \frac{\beta(1-\beta^T)}{1-\beta}\left(\sum_{s=1}^{S}\sum_{k=1}^{K^s}\frac{\omega_{s,k}E\left[\left(\widehat{\theta_t^{s,k}}-\theta^{s,k}\right)^2\right]}{(\theta^{s,k})^2}\right) \tag{3}$$

Given the fraction with β is a scaler that doesn't affect the minimum, it can be removed from the objective function, so by how much (or whether at all) to discount the variance of future estimates ends up being irrelevant to the optimal solution.

2.3 Cost Constraint

I assume that expected survey cost per case is constant within a stratum and is expressed in real (inflation-adjusted) units of currency. For the initial recruitment, the cost for a respondent is $c_{0,h}^R$, the cost for a nonrespondent is $c_{0,h}^{NR}$, and the cost for an ineligible case is $c_{0,h}^{IE}$. For each topical survey, the cost for a respondent is $c_{t,s,h}^R$ and the cost for a nonrespondent is $c_{t,s,h}^{NR}$. The total budget available is C_0 for recruitment and C_t for $t \geq 1$ to run the topicals each year. I make the simplifying assumption that the topical budget is constant over time and the costs per case are constant over time, so the relevant budgets are just C_0 and C_1. Similar to Mendelson and Elliott (2024), I have the survey practitioner make sure *expected* survey costs do not exceed the budget constraints, given that the exact costs are a random variable that is dependent on the number of realized respondents.

I could propose a maximization problem with multiple budget constraints across time periods. However, some of these constraints may not be able to be satisfied with equality simultaneously, resulting in more complex solutions. To simplify, I instead allow budget shortfalls or surpluses from initial recruitment to carry forward over time. This approach reduces the problem to a single budget constraint. While many survey organizations may be unable to adopt such carryforward processes (due to contract stipulations or government rules requiring funds to be spent within a specific timeframe) the assumption offers clear benefits. It yields a simpler solution that still incorporates time-varying costs and response rates.

If after the initial recruitment the survey practitioner has money left over, this money can grow at a real interest rate of π. If the survey practitioner instead has a shortfall after initial recruitment, then π is the interest rate for the debt. I assume then over the course of T years, the survey practitioner expends the surplus at a constant value for each year or pays off the debt at a constant rate. This amount can be derived from the standard annuity formula.

From this setup, I specify the annual budget constraint as

$$C_1 + A = \sum_{h=1}^{H}\sum_{s=1}^{S} n_h^I \phi_h(1-\psi_h)\xi_{s,h}\phi'_{s,h}\, c_{1,s,h}^R \;+\; n_h^I \phi_h(1-\psi_h)\xi_{s,h}(1-\phi'_{s,h})\, c_{1,s,h}^{NR} \tag{4}$$

where:

$$A = \left(C_o - \sum_{h=1}^{H} n_h^I \phi_h(1-\psi_h)c_h^R \;+\; n_h^I(1-\phi_h)(1-\psi_h)c_h^{NR} \;+\; n_h^I \psi_h c_h^{IE}\right)\frac{\pi}{1-(1+\pi)^{-T}} \tag{5}$$

In Equation 5, the term in the parenthesis is the budget surplus or shortfall from the recruitment stage, and the fraction with π is the annuitization factor. Note that this function is linear with respect to n_h^I. To rearrange equation (4) to highlight the linearity with respect to the initial recruitment sample size, let $\lambda = \frac{\pi}{1-(1+\pi)^{-T}}$ and

$$\alpha_h = \lambda\big(\phi_h(1-\psi_h)c_h^R + (1-\phi_h)(1-\psi_h)c_h^{NR} + \psi_h c_h^{IE}\big) + \phi_h(1-\psi_h)\sum_{s=1}^{S}\xi_{s,h}\phi'_{s,h}\, c_{1,s,h}^R + \xi_{s,h}(1-\phi'_{s,h})\, c_{1,s,h}^{NR} \quad (6)$$

The α_h terms represent the total per-unit cost of sampling one address from stratum h for the baseline survey. The total costs include the expected cost for the baseline plus the discounted costs for the future total surveys. With using the λ and α_h terms, the cost constraint is:

$$C_1 + \lambda C_o = \sum_{h=1}^{H} n_h^I \alpha_h \quad (7)$$

2.4 Final Objective Function

Given this, the final objective function is:

$$\min_{\{n_1^s,\ldots,n_H^s\}} \frac{\beta(1-\beta^T)}{1-\beta}\left(\sum_{s=1}^{S}\sum_{k=1}^{K^s}\frac{\omega_{s,k}E\left[\left(\widehat{\theta_t^{s,k}}-\theta^{s,k}\right)^2\right]}{(\theta^{s,k})^2}\right) \quad (8)$$

Subject to

$$C_1 + \lambda C_o = \sum_{h=1}^{H} n_h^I \alpha_h$$

Given 1) the term with β is a constant that doesn't affect the optimal values of the initial sample sizes and 2) the cost constraint is linear with respect to the initial sample sizes, this objective function is largely similar to that of a setup for a cross-sectional survey. The main difference is a more complicated cost constraint that includes a discount factor, and costs that are broken up between the initial recruitment costs and costs for each topical survey.

The last remaining step is to express $E\left[\left(\widehat{\theta_t^{s,k}} - \theta^{s,k}\right)^2\right]$ in a more tractable format, which I do by following logic similar to Mendelson and Elliott (2024). If the estimator is unbiased, then the expression $\mathrm{E}\left[\left(\widehat{\theta_t^{s,k}} - \theta^{s,k}\right)^2\right]/\left(\theta^{s,k}\right)^2$ equals the relative variance of the estimator $V(\widehat{\theta_t^{s,k}})\,/\left(\theta^{s,k}\right)^2$, which is the objective function from Valliant et al. (2018). If variable k is not a domain mean, then the estimator is in fact unbiased. For domain means, the estimator is biased due to the division by $D_{t,h}^{s,k}$ in the estimator, which is a random variable rather than a constant. However, the estimator is approximately unbiased if the sample sizes are large, or the domain size is large as well (Cochran 1977). Given this, I replace $E\left[\left(\widehat{\theta_t^{s,k}} - \theta^{s,k}\right)^2\right]$ by $V(\widehat{\theta_t^{s,k}})\,/\left(\theta^{s,k}\right)^2$ given the two are equal or approximately equal in a large set of cases.

Next, by the total law of variance, I can decompose $V(\widehat{\theta_t^{s,k}})$ as

$$V\left(\widehat{\theta_t^{s,k}}\right) = E\left(V\left(\widehat{\theta_t^{s,k}}\middle| D_{t,h}^{s,k}\right)\right) + V\left(E\left(\widehat{\theta^{s,k}}\middle| D_{t,h}^{s,k}\right)\right) \tag{9}$$

Because $E(\theta_t^{s,k})$ does not depend on $D_{t,h}^{s,k}$, the right-most term is zero. For $V\left(\widehat{\theta^k}\middle| D_h^k\right)$, this can be expressed as

$$V\left(\widehat{\theta_t^{s,k}}\middle| D_{t,h}^{s,k}\right) = \sum_{h=1}^{H}\left(\frac{M_h^{s,k}}{M^{s,k}}\right)^2\left(1 - \frac{D_{t,h}^{s,k}}{M^{s,k}}\right)\frac{\left(\sigma_h^{s,k}\right)^2}{D_{t,h}^{s,k}} \tag{10}$$

Which is the weighted average of the variance of the mean for a given sample size. In this equation, $\left(1 - D_{t,h}^{s,k}/M^{s,k}\right)$ is the finite population correction. Therefore, I have

$$V(\widehat{\theta^k}) = \int \dots \int \sum_{h=1}^{H}\left(\frac{M_h^{s,k}}{M^{s,k}}\right)^2\left(1 - \frac{D_{t,h}^{s,k}}{M^{s,k}}\right)\frac{\left(\sigma_h^k\right)^2}{D_h^k}\, df\left(D_1^{s,k}, \dots, D_H^{s,k}\middle| n_1^I, \dots, n_H^I\right) \tag{11}$$

where $df(.)$ Represents the probability distribution of the topical domain sample sizes conditional on the number of cases sampled for the initial recruitment. Computationally, this integral can be computed via simulation, by drawing a number of respondents, and a number of respondents in a domain, conditional on a given sample size.

In Appendix A.1, I present details on how I program this model. This model is programmed in Python using SciPy library for minimization routines.

2.5 Model Discussion

The benefit of this model is that it provides a sampling rate solution for online panels that explicitly takes into account response rate and survey cost information both for the initial recruitment and each topical survey. The simplifying assumptions I employ allow the model to largely resemble the sampling problem

for a cross-sectional survey, which makes the model easier to solve numerically. However, some of the assumptions remove common features of online panels. First, the number of topical surveys for online panels often changes from year to year. This limitation shouldn't be too problematic as long as the list of topicals used in the model is representative of the types of topicals fielded in a year.

More limiting is the assumption that the probability of response is constant over time, which removes the possibility of attriting before the end of the panel. If stratum with higher attrition rates also have lower recruitment rates and lower initial response rates to topical surveys, then ignoring attrition may not be too problematic. However, if the patterns of attrition differ from the patterns for initial recruitment, then ignoring attrition in the model may be more problematic. In this case, the objective function would largely be in the same form, but the discount factor would become relevant as later topicals surveys with different response rates would receive a higher discount. Similarly, the assumption that survey responses are independent over time is a limitation, as individual responses are likely correlated across time. This temporal correlation may influence the optimal sampling rates for cross-sectional estimates. Moreover, longitudinal analyses that examine changes in individual responses over time may be of interest in many online panel settings.

Additionally, replenishment samples are often important for online panels. Appendix A.1 discusses how the algorithm has the option to incorporate replenishment samples. In this design, the number of existing respondents is treated as a set of free cases that exist at time zero, which are then fed into the calculations for the number of new cases to sample. While this modification is somewhat artificial and doesn't fully capture how replenishment samples may be drawn repeatedly over time, it at least provides some framework to apply this model in a replenishment sample setting.

3. Data

To apply this model to a survey, I use the U.S. Census Bureau's Census Household Panel (CHP). This panel survey was launched in 2023. The design consisted of an initial "baseline" survey during the recruitment phase, followed by monthly "topical" surveys sent to all respondents. Having all panelists be in scope for each topical survey is somewhat unique to the CHP, since most other non-government panels sample a subset of panelists for each topical survey. The purpose of this panel was to collect timely information of economic and demographic topics, collect data for the Census Barriers, Attitudes and Motivators Study (CBMAS), as well as to provide a platform to test content from other Census Bureau surveys that are transitioning to add an internet self-response option. As of 2025, the CHP has been merged with the Household Pulse Survey to form a new panel called the Household Trends and Outlook Pulse Survey (HTOPS). For this paper, I use data from the initial sample and baseline survey for the CHP and its second topical survey, which were fielded in September-December 2023. See Appendix A.3 for the Preferred Reporting Items for Complex Sample Survey Analysis (PRICSSA) checklist from Seidenberg et al. (2023) for more details about the CHP. Further details about HTOPS can be found in U.S. Census Bureau (2025), in which many of the technical details about HTOPS apply to the CHP as well.

One key innovation of the CHP (and now HTOPS) is the use of the Census Bureau's Demographic Frame for sampling. This dataset harmonizes past decennial census data, federal administrative data, state administrative data, and commercial records to create a best estimate of where an individual is living in the Unites States. The CHP/HTOPS uses the Demographic Frame to flag households that likely have individuals who are Hispanic or non-White. If an address contains no data in the Demographic Frame,

then geographic estimates from the American Community Survey are used instead to assign race and ethnicity of the address.

The CHP has a multi-stage sampling design. The CHP initially stratifies the U.S. by the four Census Regions. Within Census Region, the flagged addresses are sampled at twice the rate of addresses that primarily have White non-Hispanic individuals. This oversampling rate was determined based on a rule of thumb using some rough sample size calculations and relative response rate information from past surveys. Given the limited information used to determine the sampling rates, further refinements may lead to improved precision of estimates.

For the CHP, panelists were recruited with a mailed invitation letter. Sampled households had the opportunity to respond via the web or by calling into a call center. An outbound Computer-Assisted Telephone Interview (CATI) operation followed-up with select nonresponding sampled cases as well. Once recruited, all panelists were sent survey invitations to monthly topical surveys. These survey invitations were sent out via email and text messaging. Panelists could also continue to respond via the web or CATI, either by initiative themselves to call, or via an outbound CATI operation to reach select nonrespondents.

4. Results

4.1 Model Application

To apply this model to the CHP, I make various simplifications to help with the construction of key parameters. First, I ignore the stratification by region and only consider the race and ethnicity stratification step. Second, the strata I analyze are race and ethnicity categories of 1) White non-Hispanic 2) Black non-Hispanic (abbreviated Black henceforth) 3) Hispanic and 4) All other race groups, non-Hispanic (abbreviated Other Race henceforth).[2] This collapses some categories in the original data. The frequency of addresses on the frame that are in each Strata is given in Table 1

Table 1: Strata Proportions

Frame Strata	Population Proportion
1: White Non-Hispanic	63.69%
2: Black	12.19%
3: Hispanic	14.66%
4: Other Race	9.46%

Source: Census Household Panel

Second, the only costs analyzed in the paper are costs from mailing and phone call operations. While there was a cost to the email and text messaging operations, this cost was primarily a fixed cost (i.e., independent of $n_1^s, \ldots, n_H^s$) that is not relevant to the sample allocation problem, which only considers

[2] In the subsequent analyses, I drop a small number of cases that were in a stratum that represented missing race or ethnicity. This was 1.593% of the sample frame.

variable costs. Full details on how survey costs are assigned are available in Appendix A.2. Third, I assume the unit-level variances $\sigma_h^{s,k}$ of the outcome variables are constant across the strata for all variables. This helps simplify the optimization problem to focusing mainly on response rate and cost differences.

Third, there was only one survey outcome of interest, and the variance of this variable is constant across strata. However, I assume subgroup analyses are of interest. My primary specification consists of analyzing an arbitrary variable for 1) the overall national population 2) White Non-Hispanics and 3) Hispanics and non-Whites. Race-ethnicity is measured by the primary respondent's race and ethnicity in the baseline survey. I then calculate the percent of households that are White Non-Hispanic in each stratum to assign domain frequencies. Table 2 presents these frequencies. While it is clear that the race-ethnicity category on the frame is correlated with race-ethnicity reported in the survey, there are differences. Reasons for these differences include the administrative data having a lag compared to the reported survey data, possibly resulting in misassignment of people to places, and errors from assigning race-ethnicity for some cases based on only geographic-level race-ethnicity statistics.

Table 2: Domain Proportions

Frame Strata	Percent White Non-Hispanic in Baseline Survey
1: White non-Hispanic	87.37%
2: Black	15.06%
3: Hispanic	26.76%
4: Other Race	32.88%

Source: Census Household Panel

4.2 Response Rate Differences

To help provide some intuition on upcoming results, I first present response rate and mode difference by stratum. For cross-sectional surveys, Mendelson and Elliott (2024) show that strata will have higher optimal sampling rates the lower their response rates are relative to other strata and the lower the per-unit costs are relative to other strata. The intuition here is that if a stratum has a lower response rate, then more cases will need to be sampled in order to have a sufficient number of respondents at the end of data collection. And if a stratum has lower costs, then this stratum could be oversampled to some degree in order to get a higher total number of respondents for a lower cost.

Table 3: Response Rate and Mode Differences

Frame Strata	Response Rate Baseline	Response Rate Topical	Percent Baseline Response via CATI
1: White non-Hispanic	18.97%	59.72%	9.70%
	(0.211)	(0.606)	(0.365)

2: Black	13.70%	46.56%	16.89%
	(0.300)	(1.178)	(0.883)
3: Hispanic	13.48%	45.59%	8.31%
	(0.275)	(1.092)	(0.603)
4: Other Race	18.72%	49.84%	5.18%
	(0.390)	(1.154)	(0.513)

Source: Census Household Panel. Topical response rate is the 2nd topical to the CHP, which was a set of questions from the Household Pulse Survey. Standard Errors in parentheses.

Table 3 shows CHP response rate for the baseline, the 2nd topical survey, and the percent of responses to the baseline that were through CATI. The CATI rate is presented to provide some information about costs, since cases where that respond via a phone call with an interviewer incur more of a cost than cases that self-respond online. The CATI rates impact the relative cost per stratum, which then affects the optimum sampling rates.

The table shows that the response rates of the Black and Hispanic strata are lower than the White Non-Hispanic and Other Race strata. Additionally, the Black stratum had the highest percent of response to the baseline that were in CATI at 16.89%. As a reminder, Strata 2-4 have sampling rates twice that of Stratum 1 in the implementation of the survey. Using the intuition from a cross-sectional survey, the results from Table 3 suggest that the Other Race stratum should have a lower sampling rate than Black and Hispanic Strata, given its higher response rates. Additionally, the sampling rate for the Hispanic stratum should be potentially higher than the Black stratum, given the Black stratum has a CATI rate that is about twice that of the Hispanic stratum. The next subsection explores whether these conjectures materialize in the mathematical solution.

4.3 Main Results

In this section, I present results for the optimal sampling rates for each stratum. The results are expressed in terms of sampling rates for Strata 2-4 relative to Stratum 1. A sampling rate of 2 here would match the sampling rate from the production procedures; it therefore serves as a reference for assessing the cost and efficiency gains from optimal sampling. To quantify the improvements from the optimal sampling rates over the prior sampling rates, I present two metrics. One represents the percent change in the variance of the estimates, holding the budget constant. To calculate the variance of the estimates under a given sampling rate, what I do is:

1. Using simulation, generate the number of respondents for each stratum in each topical survey, given the pre-specified budget and sampling rate per stratum. This generates a sample of respondents per each simulation. Note that while the initial sample size of the baseline survey is fixed given a budget and sampling rate, the number of respondents for the topical surveys are random variables.
2. Use these simulated numbers of respondents, calculate the variance of each survey's outcome of interest.

3. Weight each of these estimate-specific variances by $\omega_{s,k}$, the same importance weights from the optimization algorithm, to get the overall average variance, which is one number.

The percent change in variances is then the relative difference between the average variances using the optimal sampling rates compared to the average variance using the prior sampling rate (which are a relative sampling rate of 2 for Strata 2-4). Note that changing $\omega_{s,k}$ not only affects the sampling rates, but also how the variance of each outcome is weighted in the summary variance calculations.

The other metric represents cost savings, holding precision constant. This metric represents taking the actual initial sampling rates and finding the budget that would result in the same average precision of estimates as one would have gotten with the optimal sampling rates using the current budget. Numerical optimization is used to find this equivalent budget.

For example, suppose that, for a budget of $40,000, one could use the optimal sampling rate methodology to estimate a mean of a binary variable with an average variance of 2%, calculated using steps 1-3 described above. Additionally, suppose that from using a prior rule-of-thumb method to assign the sampling rates, one could get the same variance of 2% for a budget of $50,000. Note that the latter allocation has a higher budget because the sample is being allocated in a less efficient manner. This would represent a cost savings of $10,000/$50,000 = 20%. Note that these two metrics should result in similar relative changes, but the exact numbers may differ slightly given the comparison variance is calculated using different budgets and sampling rates.

Table 4: Sampling Rates with Varying Response and Cost Parameters

	Relative Sampling Rates				
Specification	Stratum 2: Black	Stratum 3: Hispanic	Stratum 4: Other Race	Percent Change Variance	Percent Change Cost
Response Rates and Costs Equal	1.265	1.284	1.251	-3.27%	-3.27%
Costs Equal	2.43	2.381	1.502	-2.58%	-2.27%
Response Rates Equal	1.275	1.29	1.439	-4.04%	-3.62%
No Equality Restrictions	2.363	2.439	1.636	-1.69%	-1.46%

Source: Census Household Panel

Table 4 presents optimal sampling rates under the specification where the overall national mean is given an importance weight of 0.6, the mean for White non-Hispanics is given an importance weight of 0.2, and the mean for Hispanic and non-Whites is given an importance weight of 0.2. To help explain how the algorithm arrives at the desired solution, I calculate the sampling rates using either the actual estimated parameters for the model, or from holding certain parameters constant across strata by assigning the estimate for the 1st stratum to the value for all other strata. The first row shows the result from holding both response rates and cost parameters constant. Under this scenario, the oversampling rates are between 1.251 and 1.284 for Strata 2-4. Under this scenario, these strata are still oversampled to some degree in order to have an adequate number of respondents for the Hispanic/non-White mean. Thus, because of the domain analysis, the optimal sampling rates are still not 1, which would be the Neyman (1934) allocation.

The next two rows show effects of holding cost equal or holding response rates equal. The largest change in the sampling rates comes from having the response rates differ between strata (the cost equal row). However, allowing only costs to differ also results in a higher sampling rate for Strata 2-4 relative to Stratum 1. Finally, when removing all equality conditions and using the full set of estimated parameters, the sampling rates for the Black and Hispanic strata are somewhat higher than the value of 2 from the production implementation, at 2.363 and 2.439, respectively. The Other Race stratum has an optimal sampling rate of 1.636 which is lower than the production specification. This modest change in sampling rates results in a small amount of cost-savings. The model results show that the optimal sampling rates would reduce average variance by 1.69%, which is equivalent to obtaining the same average variances with a budget that is 1.46% lower.

4.3 Sensitivity Analysis

Next, I conduct some sensitivity analyses to see how the sampling rates change with different prioritizations. Table 5 shows how the sampling rates change with different importance weights. The importance weight for the national estimate of 0.6 corresponds to the solution in the prior table. Overall, the results do not appear to be too sensitive to the choice of importance weights. When the national estimate gets an importance weight of 1/3, the sampling rate for the Black stratum is 2.627. The sampling rate drops to only 2.299 when the national estimate gets an importance weight of 2/3.

Table 5: Change in Importance Weights

Importance Weight			Relative Sampling Rates				
1: National	2: White non-Hispanic	3: Hispanic/Non-White	Stratum 2: Black	Stratum 3: Hispanic	Stratum 4: Other Race	Percent Change Variance	Percent Change Cost
0.333	0.333	0.333	2.627	2.556	1.74	-2.62%	-2.31%
0.500	0.250	0.250	2.577	2.633	1.711	-2.02%	-1.67%
0.600	0.200	0.200	2.363	2.439	1.636	-1.69%	-1.46%
0.667	0.167	0.167	2.299	2.324	1.558	-1.47%	-1.19%

Source: Census Household Panel.

The estimates change more when I change the subgroup analysis to split the Hispanic/non-White group into 3 race and ethnicity categories. This creates a sampling analysis which considers the precision of 5 means. Table 6 shows the results from this analysis. Under this framework, more of the sample is being allocated to Strata 2-4. The sample rate for the Other Race stratum stays below 1.45. The Hispanic sampling rate is higher than the Black sampling rate. When the national estimate is given an importance weight of 0.6, the Black stratum has a sampling rate of 3.835, compared to 5.148 sampling rate for the Hispanic stratum. These differences are potentially due to the lower CATI rate for the Hispanic stratum (lower CATI rates are associated with lower cost per case). If a survey practitioner decides this specification for subgroup means is desirable, then these optimal sampling rates result in higher cost savings. Cost savings range from 11.92% to 18.66% depending on the value of the importance weights. Do note that while the importance weights in Table 6 give similar weight to the national estimate as in Table 5, the importance weight for the White non-Hispanic group is greatly reduced. Thus, the lower implied sampling rate for the White non-Hispanic group here may be due to both having more race-

ethnicity domain means as well as a lower importance weight being assigned to the White non-Hispanic group.

Table 6: Additional Domains

Importance Weight					Relative Sampling Rates				
1: National	2: White non-Hispanic	3: Black	4: Hispanic	5: Other Race	Stratum 2: Black	Stratum 3: Hispanic	Stratum 4: Other Race	Percent Change Variance	Percent Change Cost
0.500	0.125	0.125	0.125	0.125	4.34	5.847	1.391	-17.25%	-18.66%
0.600	0.100	0.100	0.100	0.100	3.835	5.148	1.303	-15.01%	-16.00%
0.750	0.063	0.063	0.063	0.063	3.444	4.131	1.45	-10.87%	-11.40%
0.800	0.050	0.050	0.050	0.050	2.828	3.738	1.338	-9.27%	-11.92%

Source: Census Household Panel.

5. Conclusion

This paper shows how to assign sampling rates in online panels for stratified samples. This approach offers benefits over either rule-of-thumb assignment of sampling rates or using sampling rate formulas from the 20th century. To assess the performance of the optimal sampling formula I develop, I apply this model to the Census Household Panel. The benefits of these new optimal sampling rates vary depending on the specification. When the subgroup analyses are limited to comparing means of White non-Hispanics to all other groups combined, the sampling rates are similar to the production implementation, and the cost savings are only about 1%. If instead the survey practitioner cares about the precision of means for the four race and ethnicity groups separately, then the optimal sampling solution results in higher sampling rates for the Black and Hispanic strata. In this case, the cost-savings amount to 11.92% to 18.66%, depending on the value of the importance weights. This shows that the sampling solution can change significantly depending on how one specifies the survey priorities.

The presented model attempts to strike a balance between operational feasibility and incorporating relevant features of online panels. One feature that is left out is panel attrition, which is left to future research. The importance of panel attrition may depend on how comparable the strata-specific attrition rates are to initial baseline response rates and topical response rates. If the strata with higher attrition rates also have lower response rates generally, then it is possible the solution proposed by this current model may be similar to a more complicated model that incorporates attrition.

Additionally, the model developed in this paper primarily considers a situation of stratification based on microdata, where the population is divided into a small number of strata (e.g. less than 10). Many surveys have multi-stage sampling strategies that sample geographies first (such as counties in the United Staties) followed by sampling households within selected geographies. The model presented in this paper focuses on the second part of sampling households or individuals. One can still use this model to apply sampling rates in a multi-stage setting by assigned the same second-stage sampling rate with each geography based on the national-level model presented in this paper. However, this is likely not optimal, given the distribution of characteristics usually varies by geography. For example, if household with non-White or Hispanic individuals are to be oversampled, assigned a higher sampling

rate to counties with a higher portion of such individuals helps reach this goal to some degree, which may then affect what the sampling rates should be within geography. Future research can explore how to set sampling rates for online panels in multi-stage designs.

Going beyond the implications just for online panels, these results help advance the literature for survey sampling generally in the presence of nonresponse. The framework in this paper could be modified to longitudinal surveys, using similar modifications to express the sampling objective function in a way that resembles the objective function for a cross-sectional survey. Additionally, while this is not the first paper to use numerical optimization to calculate sampling rates, the application of numerical optimization to sampling still appears to be limited. Some details on the numerical optimization methods and details on how domain means are handled in the model, for example, could serve as useful tools for other sampling problems that aren't confined to online panels.

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Appendix

A.1. Computational Details

Implementation

I have developed a python program called "optimal_sampling_online_panel" that estimates the model described in Section 2. It uses the minimization routine in the SciPy package to solve the objective function specified in equation 8, where the variance of the estimates are as specified in equation 11. The python code is available at the U.S. Census Bureau's GitHub page at https://github.com/uscensusbureau/optimal-stratified-sampling-with-python/, and the code documentation describes how to use the function. The following are some notes about the implementation of the model that are relevant for understanding the program's characteristics.

1. **Simulation**. I use simulation to evaluate the integral in equation 11. The potential number of respondents in each simulated sample is fixed at the beginning of the program, using the response rate parameters to simulate whether an initial sampled case responds to the baseline, and whether someone recruited for the panel responds to a given topical survey. So, for example, if the current guess of the optimal solution is 2,000 sampled cases for a stratum, then I take the first 2,000 rows from a pre-defined vector of response indicators to determine the number of respondents in the baseline and the topical survey in this simulated sample. Similar procedures are performed to figure out the number of respondents in a domain. This is done in order to fix the random number for each iteration of the minimization routine. It is well-known that procedures like this are needed in minimization with simulation; if new sets of respondents were drawn for each iteration of the minimization routine, then the objective function would oscillate and never converge.
2. **Continuous Transformation of Sample Counts.** The model estimates the number of respondents, and number of respondents in a domain, as continuous variables, given the programs take continuous variables as inputs. To augment what is described above, consider a stratum with a response rate of 62.5% to the baseline survey and a response rate of 80% to a topical survey, resulting in a cumulative response rate of 50%. Take a simulated sample where, for 2,000 sampled cases, the realized number of respondents to the topical is 1,050. If the current guess of the optimal solution is 2,000.3 sampled cases, then I multiply the remainder by the cumulative response rate and add that to the integer-value for the simulated sample sizes. From the prior numerical example, the realized number of respondents would become 1,050 + .3*.5 = 1,050.15. This makes the simulated estimator continuous with respect to the input variables, which is needed for derivative-based maximization routines.
3. **Unconstrained Optimization.** The model inputs are the number of sampled observations for each stratum, minus the count for the very last stratum. The number of observations for the last stratum is calculated based on the amount of money left over in the expected cost after calculating the expected cost from fielding the other strata. Thus, for each new guess of the optimal number of sampled cases, the algorithm:
    a. Takes the current guess of the number of samples cases (not including the last stratum) and calculates the expected costs for stratum 1 through stratum $H-1$.
    b. Takes the difference between this and the budget to get the expected amount of money spent on strata H.
    c. From this amount of money, calculate the initial sample size for strata H that would result in this expected cost.

    This procedure then transforms the problem from a constrained minimization to an unconstrained minimization problem. I do this to help with model performance and to increase the number of minimization algorithms I can use from the python library. For the algorithm, I ensure that the initial guess is feasible and wouldn't result in a negative sample size for the last Strata. I also introduce a large penalty value into the objective function if any of the initial guessed strata sizes ever become negative. While this introduces a discontinuity in the objective function, this adjustment seemed to work well to have the optimization algorithm stick to feasible values.
4. **Global Optimization:** I use the Basin-hopping global optimization procedures combined with the L-BFGS-B algorithm for the local minimizer. This procedure involves finding the minimum

with the local minimizer, perturbing the solution, and then running the local minimization again to see if the solution has improved. Using the global minimizer appears to improve the performance of the optimization process and reduce the sensitivity of the solution to initial values. While the continuous transformation of sample counts does smooth the simulator, testing suggested that the local minimizer by itself would sometimes fail to find the global minimum.

5. **Parameters for Numerical Derivatives.** The program has default parameters for the minimization algorithm to use, the step-size for calculating numerical derivatives (eps), and a parameter for scaling the objective function (scale_factor). These parameters were needed because initial testing determined that if the step size was too small or the objective function returned too small of a value, the minimization routine would not return the optimal solution. If one is having convergence issues with this program, they should modify these parameters to see if this affects the returned solution.
6. **Initial Values:** For each sampling rate presented in the paper, the algorithm was run under two starting conditions of 1) the production sample sizes (where the Hispanic and Non-White strata are sampled at the doubled rate) and 2) one where most of the sampled cases were assigned to the last strata. The solutions from the two starting values were compared to ensure that they were approximately equal.

A.2 Estimation of Model Parameters

The model parameters are all estimated from the Census Household Panel for data collected between September 2023 and December 2023. Below is a description of how the parameters are estimated:

Ineligibility Rates

The CHP does not have any in-person interviews to confirm if the sampled address was actually an occupied housing unit. Because of this, I use an internal-use version of the American Community Survey. For each address sampled in the ACS that year, I calculated the percent of addresses that were vacant, commercial, or non-existent. I then apply this rate to the CHP strata to get the ineligibility rate.

Response Rates

The baseline response rate comes from the initial baseline survey for CHP fielded in September and October of 2023. Response rates are AAPOR RR3, using the ineligibility rate described above. Base weights are used to construct this response rate. Conditional on being a panelist, response rates for the topical survey were calculated from Topical 2, which was fielded in December 2023. Response rates here are AAPOR RR6, using the base weight as well.

Baseline Cost Estimates

For the baseline survey, up to three postal mailings were sent to each address. The mailing cost for one survey invitation letter was calculated using internal cost data. The initial mailing contained a $5 unconditional cash incentive. For phone costs, the calculations are more complicated. The calculations involve estimating the average number of hours a CATI interviewer spent on one case in a stratum and then multiplying this by the CATI hourly cost for the Census Bureau's CATI center based on internal cost data.

For the outbound CATI operations, certain households were assigned to receive up to two phone calls, others only one phone call, and other groups received no outbound phone calls. The groups were assigned based on the tract-level self-response rates to the American Community Survey.[3] From the tract-level self-response rates, households were divided into quartiles. Households in the 2nd quartile received two outbound phone calls and households in the third quartile were in the one outbound phone call group. Household in the 1st and 4th quartile did not receive any outbound phone calls. Using this information, for each sampling stratum, I calculate the percent of households that were in the one-phone call group and the two-phone call group. Each outbound phone call was inferred to take a set number of minutes of an interviewer's time, not including any time from an actual interview (that is, time spent asking the survey questions and recording the individual's response). The number of minutes per outbound phone call was set based on internal cost data. These statistics were then used to assign the cost to the outbound CATI operation, not including any time for actual interviews.

For the CATI interview portion, the internal data indicate whether a case responded by CATI. There is no information on whether the CATI response was from an inbound or outbound call. The start and stop times of successful CATI interviews are recorded. But this is only for when the CATI interviewer reaches a respondent, there is no case-level data on number of minutes spent on outbound phone calls that go to voicemail, for example. From the information I have available, I calculate by stratum the percent of responses that are from CATI and the average length of a CATI interview, winsorized at the 99th percentile. These two statistics give the average amount of time per case that is spent on a CATI interview. From here, total time spent per case on CATI is the time spent on outbound calls described in the prior paragraph, and the time spent on a CATI interview described in this paragraph.

Households that responded to the baseline received an additional $20 conditional incentive.

Topical Cost Estimates

For the topical surveys, there are no mailings, so there are no mailing costs. Phone costs are assigned in a similar manner as the baseline survey. The main difference between topical and baseline surveys is how the groups are assigned for the number of outbound calls. If the panelist only has a landline phone and did not provide an email address, they get up to two outbound calls. Panelists with a cell phone but no email get one outbound call. Panelists also received a $10 conditional incentive for completing a topical interview.

Time Parameters

For the time-related parameters in the model, I assume the panel will last 5 years. For the interest rate, I calculate a real interest rate for around the time the CHP baseline was fielded. First, I take the nominal 5-year U.S treasury bond interest rate from November 30th, 2023, retrieved from the Board of Governors of the Federal Reserve System [US] (2025). This nominal interest rate is 4.31%. The inflation rate of 3.1% is taken from the headline non-seasonally adjusted annual U.S. inflation rate from November 2023 (Bureau of Labor Statistics 2025). This gives a real interest rate of 1.21%. Given the model specification from this paper only has one topical, the interest rate and time parameters are transformed to monthly time intervals. Note that the value of discount factor β does not need to be set

[3] Census tracts are geographic areas that contain on average about 4,000 households. The tract-level response rates used for CHP were from the 2016-2020 ACS 5-year estimates.

given that the term appears as a scalar for the objective function, which does not affect the minimum value. If the discount factor did need to be set, one could use the Intertemporal Euler Equation of $\beta = 1/(1+r)$ to obtain the discount rate from the real interest rate.

A.3 PRICSSA Checklist

Table A.3 PRICSSA Checklist

PRICSSA item	
1.1 Data collection dates	Baseline from September-October 2023. Topical 2 from December 2023
1.2 Data collection mode(s)	Primarily Web. CATI for some respondents. Sampled households receive an initial mail invitation which gives them instructions on how to respond online or respond by calling a call-center number.
1.3 Target population	General U.S. Population
1.4 Sample design	Stratified sampled by Census Regions (which there are 4) and by race and ethnicity categories. Race and ethnicity data based on microdata from the Census Bureau's Demographic Frame (Ratcliffe 2021) and tract-level estimate from the Census Planning Database (which are themselves derived from 5-year American Community Survey Estimates). For the race and ethnicity strata, the Demographic Frame was used first if an address had any matches to the Demographic Frame. For addresses that did not contain any matches to the Demographic Frame, the tract-level estimates are used instead in assigning race and ethnicity.
1.5 Survey response rate(s)	17.4% for the baseline (AAPOR RR3). Conditional on panel participation, response rate to topical 2 was 55.8% (AAPOR RR6). Source: Hornick (2024)
2.1 Missingness rates	Hot deck imputation is used to impute missing items.
2.2 Observation deletion	No observations were deleted.
2.3 Sample sizes	Number of cases sampled for the baseline was 75,000 addresses.
2.4 Confidence intervals/standard errors	Standard errors reported in table.

2.5 Weighting	Base weights used for all response rate and cost parameter estimates
2.6 Variance estimation	For Table 3 which presents standard error of means, I used standard formulas for simple random samples.
2.7 Subpopulation analysis	Optimization code used in this paper adjusts for domain sizes being unknown ahead of time.
2.8 Suppression rules	Per Census Bureau's disclosure rules, all estimates rounded to 4 significant digits.
2.9 Software and code	Stata 18 used to clean CHP data and estimate response rate and cost-input statistics. Python used for optimal sampling code, which primary used the SciPy library, version 1.11.1. Data and code available per guidance in https://www2.census.gov/foia/ds_policies/ds027.pdf, "DS027 – Transparency in Secondary Research."
2.10 Singleton problem (as needed)	Not Applicable.
2.11 Public/restricted data (as needed)	Restricted-use data used.
2.12 Embedded experiments (as needed)	Not Applicable.

Table contains PRICSSA item checklist (Seidenberg, Moser, and West 2023).